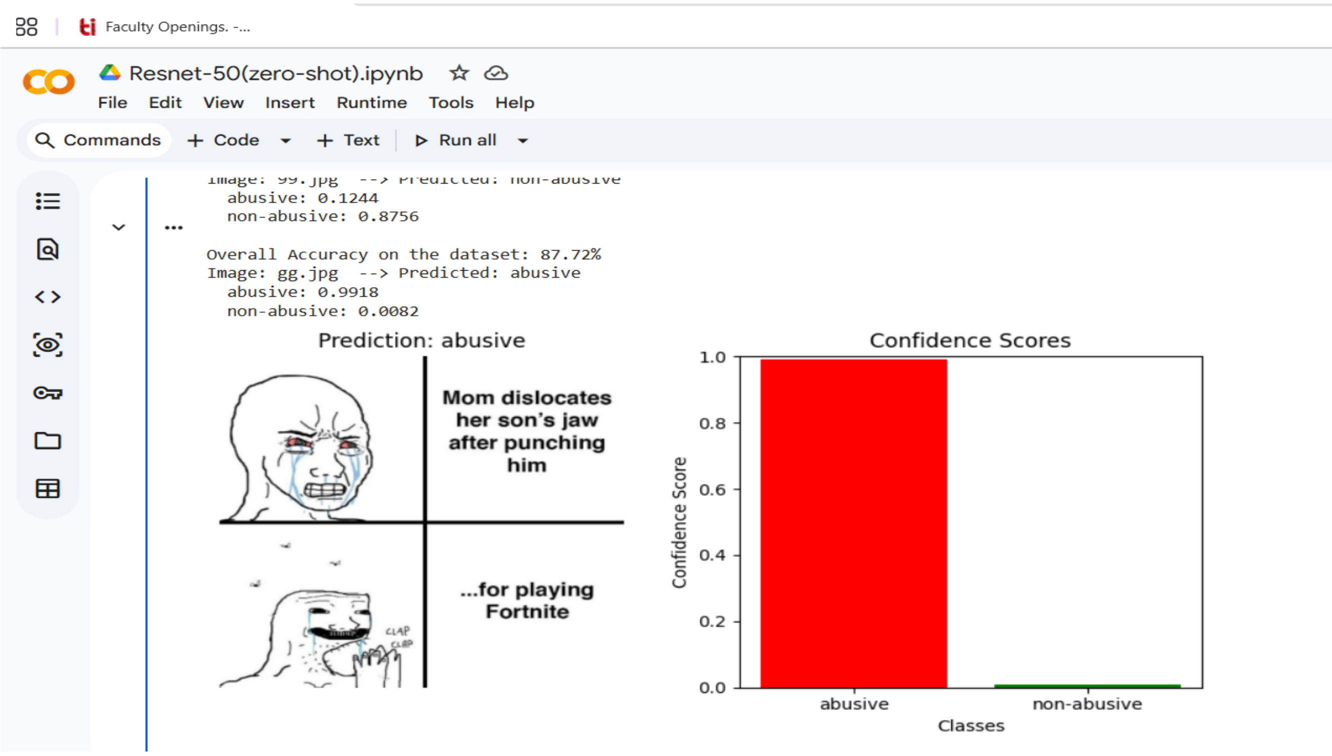
ResNet50-Based Child Abuse Meme Detection

Experimental Results: Zero-Shot, Few-Shot, and Fine-Tuned Learning

This report summarizes the performance of the ResNet50 model under three different learning settings: Zero-Shot Learning, Few-Shot Learning, and Fine-Tuned Learning. The objective was to classify memes into abusive and non-abusive categories. The experiments evaluate how performance changes as the model is provided with increasing amounts of task-specific training data.

ResNet50 Zero-Shot Learning

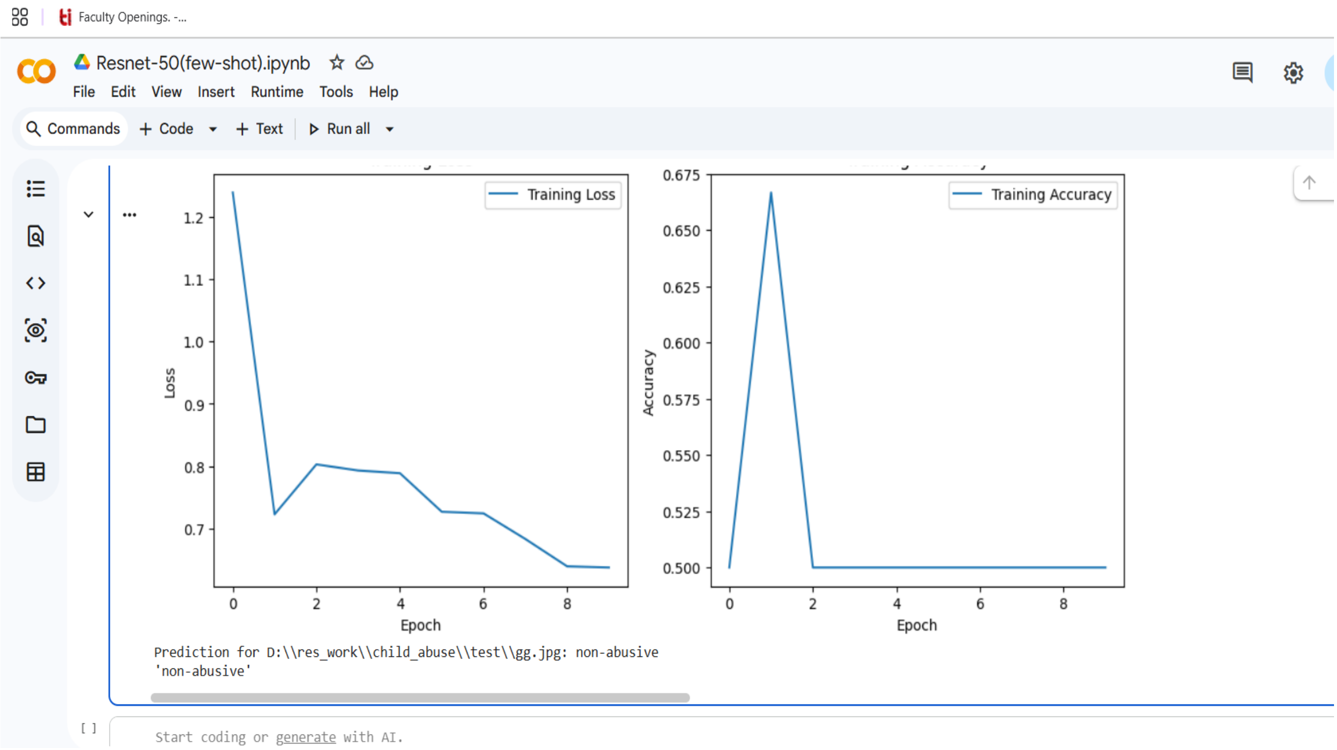


In the zero-shot setting, the model performs classification without task-specific training. The model achieved an overall accuracy of approximately 87.72% and demonstrated strong confidence in its predictions. This experiment highlights the transfer learning capability of pretrained visual representations.

ResNet50 Fine-Tuned Learning

In the fine-tuned setting, the pretrained ResNet50 model was further trained on the child abuse meme dataset. Training and validation losses decrease steadily, while validation accuracy reaches above 92%, indicating effective adaptation to the target task and improved classification performance.

ResNet50 Few-Shot Learning



In the few-shot setting, only a limited number of labeled examples were available for training. Although training loss decreases across epochs, the accuracy remains relatively low and stabilizes around 50%, suggesting that additional training samples or data augmentation techniques may be required for better generalization.

Conclusion:

Among the three settings, the fine-tuned ResNet50 model achieved the best performance and the most stable training behavior. The zero-shot model demonstrated strong transfer-learning capabilities without additional training, while the few-shot model showed the limitations of learning from a very small number of examples. These results indicate that task-specific fine-tuning significantly improves meme classification performance.